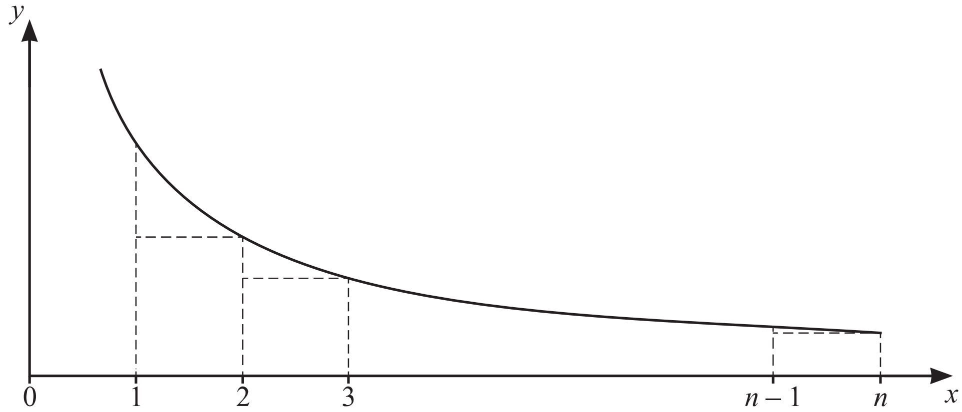
6



The diagram shows the curve $y = \dfrac{1}{\sqrt{x^2 + 2x}}$ for $x > 0$, together with a set of $(n-1)$ rectangles of unit width.

By considering the sum of the areas of these rectangles, show that

$$\sum_{r=1}^{n} \frac{1}{\sqrt{r^2 + 2r}} < \ln\left(n + 1 + \sqrt{n^2 + 2n}\right) + \tfrac{1}{3}\sqrt{3} - \ln\left(2 + \sqrt{3}\right).$$ [10]

....

....

....

....

....

....

....

....

....

....

....

....

....

....

....

....

....